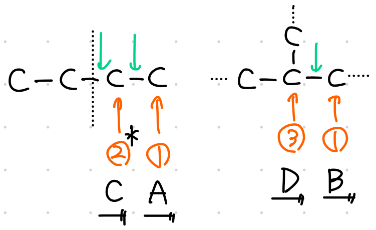
3-[1] $C_4H_{10}O$ $U=1$ { アルコール / エーテル



アルコール A① B① C* D③

Ox E F G ×

アルデヒド アルデヒド

b.p. A > B

問1

A : $CH_2-CH_2-CH_2-CH_2-OH$

B : $CH_3-CH(CH_3)-CH_2-OH$

C : $CH_3-CH_2-CH(OH)-CH_3$

D : $CH_3-C(CH_3)(OH)-CH_3$

問2 ヒドロキシ基を持ち，
分子間で水素結合を形成するため.

問3 E : $CH_3-CH_2-CH_2-C(=O)-H$

F : $CH_3-CH(CH_3)-C(=O)-H$

G : $CH_3-CH_2-C(=O)-CH_3$

C, G

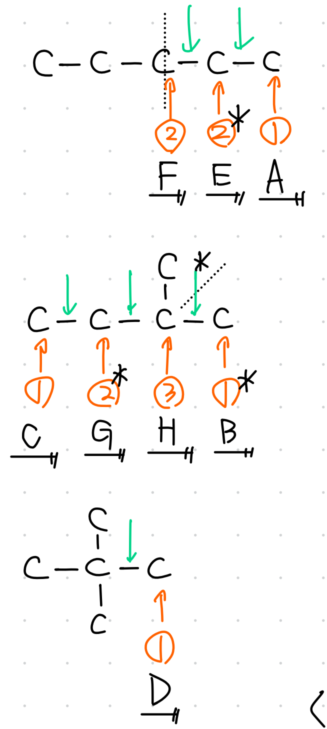
問4.

$CH_3-CH_2-O-CH_2-CH_3$

$CH_3-CH_2-CH_2-O-CH_3$

$CH_3-CH(CH_3)-O-CH_3$

3 − 2　$C_5H_{12}O$　$U=0$ { アルコール / エーテル



アルコール　①A　①B*　①C　①D　②E*　②F　②G*　③H

Ox　アルデヒド (A, B*, C, D)　ケトン (E*, F, G*)　× (H)

$I_2 + NaOH$

$-H_2O$　A′　×　F′　シス・トランスなし

同一C骨格 (A′, F′)

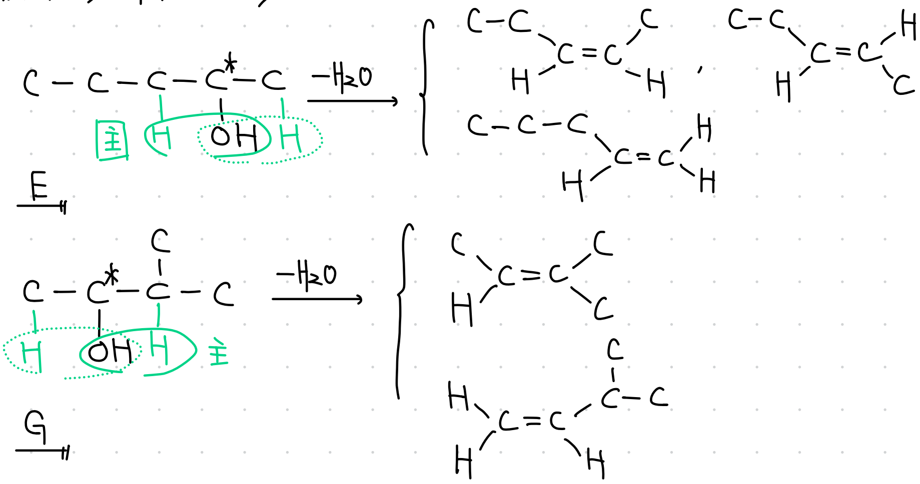
↓ $+H_2$

同一

◆「脱水して〜」「H_2付加して〜」} 同一化合物

⟶ 同一C骨格

〈2級 E*、G*について〉

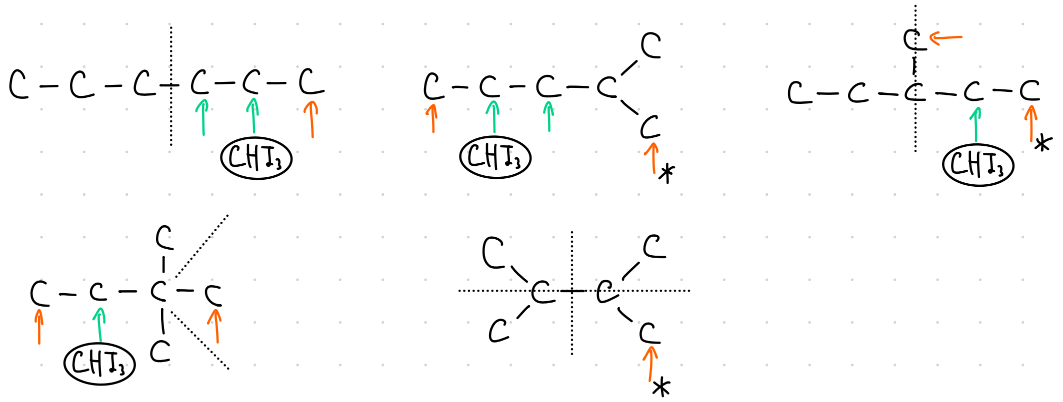


(2) 6 (立体異性：7)

3 − $\boxed{3}$

← : =O (アルデヒド)
← : =O (ケトン)

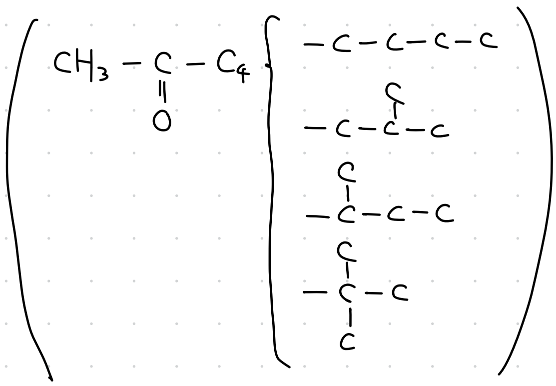
問1



(a) 8　　(b) 6

問2. 3

問3 4



問4

X　$CH_3-CH_2-CH_2-CH_2-CH_2-C(=O)-H$

Y　$CH_3-CH_2-CH_2-CH_2-CH_2-C(=O)-OH$

3 − $\boxed{4}$

問1

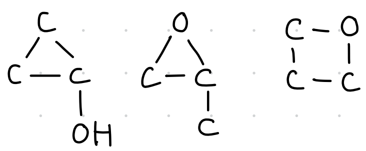
A　$CH_2=CH-CH_2-OH$

B　$CH_2=CH-O-CH_3$

C　$CH_3-CH_2-C(=O)-H$

D　$CH_3-C(=O)-CH_3$

問2



3

3 − $\boxed{5}$ $C_4H_8O_2$ $U=1$ $-\underset{\underset{O}{\|}}{C}-O-$ ×1.

$H-COO-C_3$ $\begin{cases} \boxed{D}\ -C-C-C \xrightarrow{+H_2O} \text{HCOOH (Red)} + C-C-C-OH \\ \boxed{A}\ -\underset{}{\overset{C}{\overset{|}{C}}}-C \longrightarrow \text{〃 (Red)} + C-\underset{\underset{OH}{|}}{C}-C \ (CHI_3) \end{cases}$

$\boxed{C}$ $C-COO-C_2 \longrightarrow C-COOH + C-C-OH\ (CHI_3)$

$\boxed{B}$ $C_2-COO-C \longrightarrow C-C-COOH \quad C-OH$

b.p. $C-OH < C-C-OH < \underbrace{\overset{2級}{C-\underset{\underset{OH}{|}}{C}-C} < \overset{1級}{C-C-C-OH}}_{異性体}$

(小) ⟵ 分子量 ⟶ (大)

A : $H-\underset{\underset{O}{\|}}{C}-O-\underset{\underset{CH_3}{|}}{CH}-CH_3$ ギ酸イソプロピル

B : $CH_3-CH_2-\underset{\underset{O}{\|}}{C}-O-CH_3$ プロピオン酸メチル

C : $CH_3-\underset{\underset{O}{\|}}{C}-O-CH_2-CH_3$ 酢酸エチル

D : $H-\underset{\underset{O}{\|}}{C}-O-CH_2-CH_2-CH_3$ ギ酸プロピル

3 − $\boxed{6}$

問1 $w_C = 264\,g \times \dfrac{C\,(12)}{CO_2(44)} = 72\,g$

$w_H = 90\,g \times \dfrac{2H\,(2)}{H_2O\,(18)} = 10\,g$

$w_O = 114 - (72 + 10) = 32\,g$

$C : H : O = \dfrac{72\,g}{12\,g/mol} : \dfrac{10\,g}{1.0\,g/mol} : \dfrac{32\,g}{16\,g/mol} = 3 : 5 : 1$

$(C_3H_5O)_n = 57n = 228 \quad n = 4 \qquad \underline{C_{12}H_{20}O_4}$

〈別解〉 燃焼あり, 分子量既知のとき

$C_xH_yO_z + \frac{1}{2}\left(2x + \frac{y}{2} - z\right) O_2 \longrightarrow x\,CO_2 + \frac{y}{2} H_2O$

$\underset{(288)}{C_xH_yO_z} : \underset{(44)}{CO_2} = 1 : x$ より,

$\dfrac{114\,g}{228\,g/mol} \times x = \dfrac{264\,g}{44\,g/mol} \times 1 \qquad \therefore x = 12$

$\underset{(288)}{C_xH_yO_z} : H_2O = 1 : \dfrac{y}{2}$ より

$\dfrac{114\,g}{228\,g/mol} \times \dfrac{y}{2} = \dfrac{90\,g}{18\,g/mol} \times 1 \qquad \therefore y = 20$

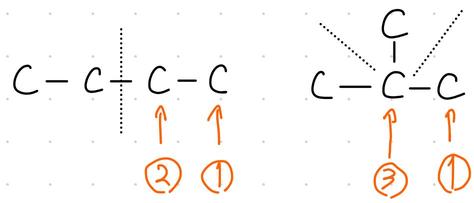
$C_{12}H_{20}O_z = 164 + 16z = 228 \quad \therefore z = 4 \qquad \underline{C_{12}H_{20}O_4}$

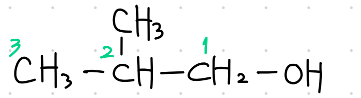
問2. A : $C_{12}H_{20}O_4$ $U = \frac{2\times12+2-20}{2} = 3$ 〰 $C=C \times 1 + -\underset{\underset{O}{\|}}{C}-O- \times 2$

ジエステルA ($C_{12}H_{20}O_4$) $\xrightarrow{\text{けん化}}$ ① アルコールB, C ($C_4H_{10}O$)
→ ②⑨ DのNa塩 $\xrightarrow{H^+}$ D

* アルコールB $\xrightarrow{Ox}$ アルデヒド
第1級
* アルコールC $\xrightarrow{Ox}$ ✕
第3級
* B, C $\xrightarrow{-H_2O}$ 同一アルケン
同一C骨格

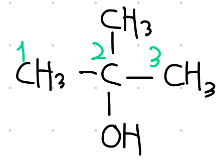
$C_4H_{10}O$ (U=0)



B : 
2−メチル−1−プロパノール

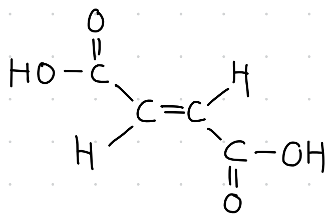
* D $\xrightarrow{-H_2O}$ ✕
↕ シス・トランス
E $\xrightarrow[-H_2O]{}$ ○

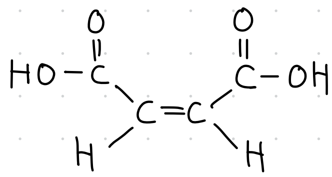
D : トランス型
E : シス型
$\underbrace{(C=C\text{あり}, -\underset{\underset{O}{\|}}{C}-OH \times 2)}_{U=3}$

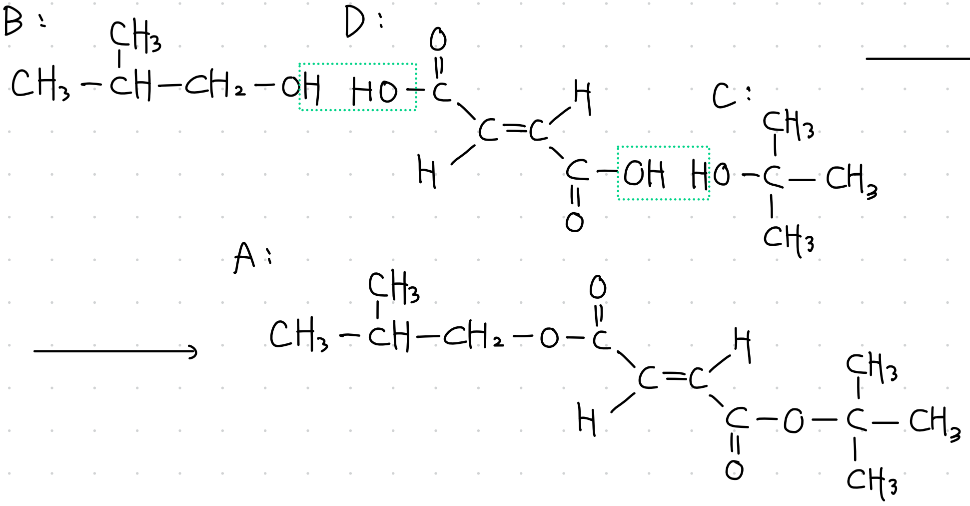
C : 
2−メチル−2−プロパノール

ジエステル
A + $2H_2O$ ⟶ B + C + D

∴ D = A + $2H_2O$ − (B+C)
= $C_{12}H_{20}O_4 + 2H_2O - 2\times C_4H_{10}O$
= $C_4H_4O_4$ (114)
(U=3)

D : 
フマル酸

E : 
マレイン酸



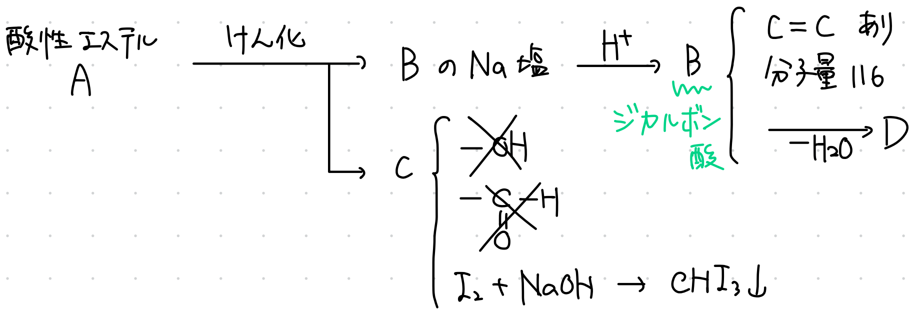
3 − $\boxed{7}$

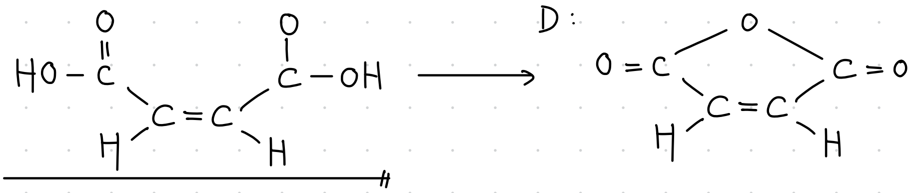
問1　$C_7H_8O_4$

$$C:H:O = \frac{53.8}{12} : \frac{5.1}{1.0} : \frac{41.1}{16} = 4.48 : 5.1 : \underset{①}{2.56}$$
$$\fallingdotseq 1.75 : 1.99 : 1 = 7 : 8 : 4$$

$130 \leqq (C_7H_8O_4)_n \leqq 170$　∴ $n = 1$.　$\underline{C_7H_8O_4}$ $(U=4)$

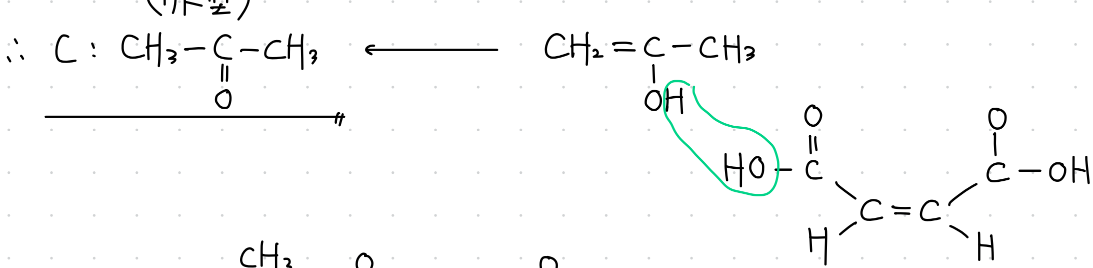
問2

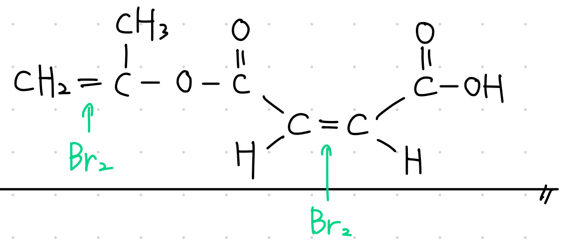


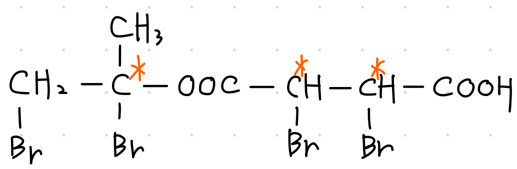
B :

D :

$A + H_2O \longrightarrow B + C$　∴ $C = A + H_2O - B$
$= C_7H_8O_4 + H_2O - C_4H_4O_4$
$= \underline{C_3H_6O}$　$(U=1)$

エステルの加水分解生成物として −OH基を持つものがない
⟶ エノール型がケト型に変わった.

(ケト型)


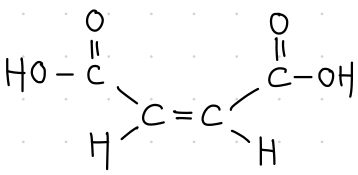
A :


問3


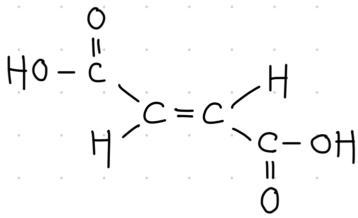
$2^3 = \underline{8}$

3 − $\boxed{8}$

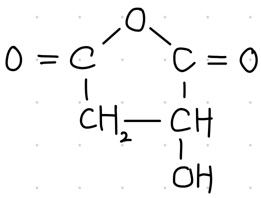
A



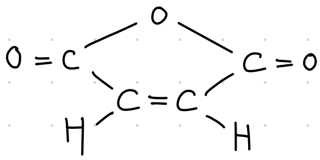
B



C



D

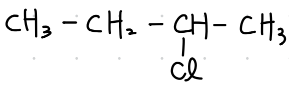


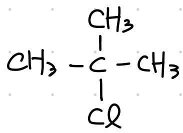
E　$HOOC-CH_2-CH_2-COOH$

3 − $\boxed{9}$

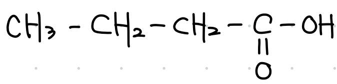
問1　4

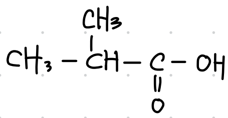
問2.



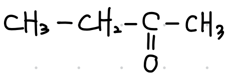


問3



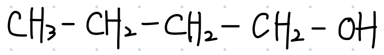


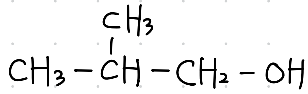
問4

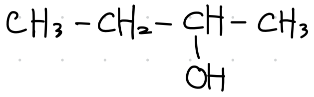


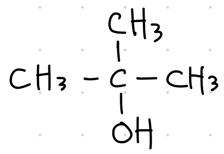
3.6g

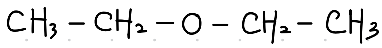
3-10

問1. A 

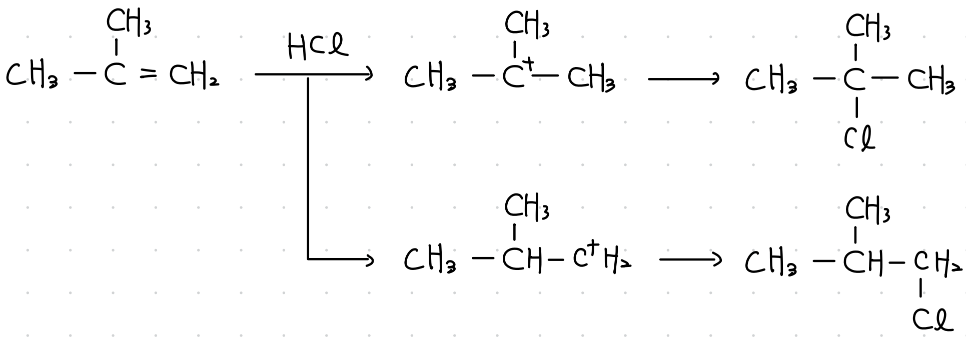
B 

C 

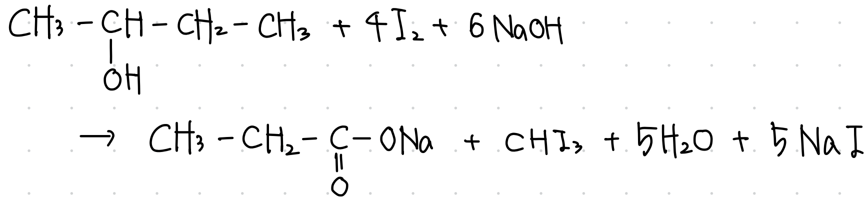
D 

E 

問2



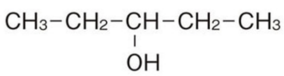
問3

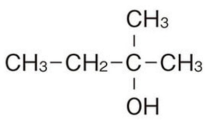


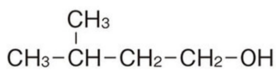
問4 C

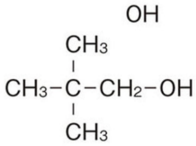
問5 エーテル結合

3-11

問1A 

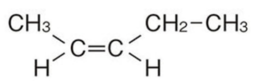
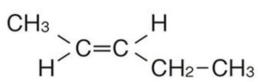
B 

C 

D 

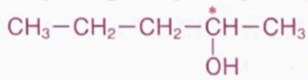
F $CH_3-CH_2-CH_2-CH_2-CH_2-OH$

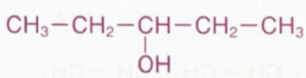
問2 K

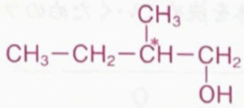
問3  

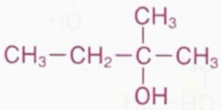
3－12

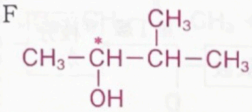
問1 A $CH_3-CH_2-CH_2-CH_2-CH_2-OH$

B 

C 

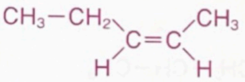
D 

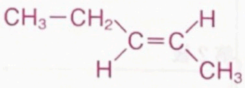
E 

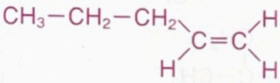
F 

問2 0.15 L　　問3 銀鏡反応，アルデヒド基

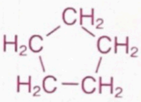
問4

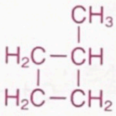


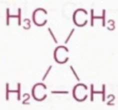


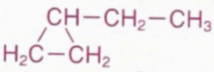


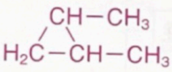
問5



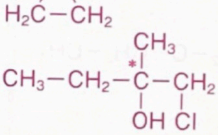


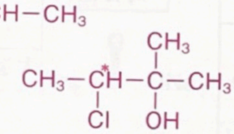






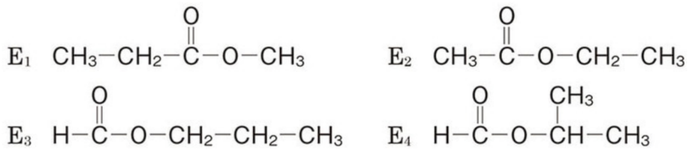
問6



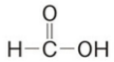


3 − $\boxed{13}$

問 1



問 2 ギ酸は分子内にホルミル基を持つため。(19 字)



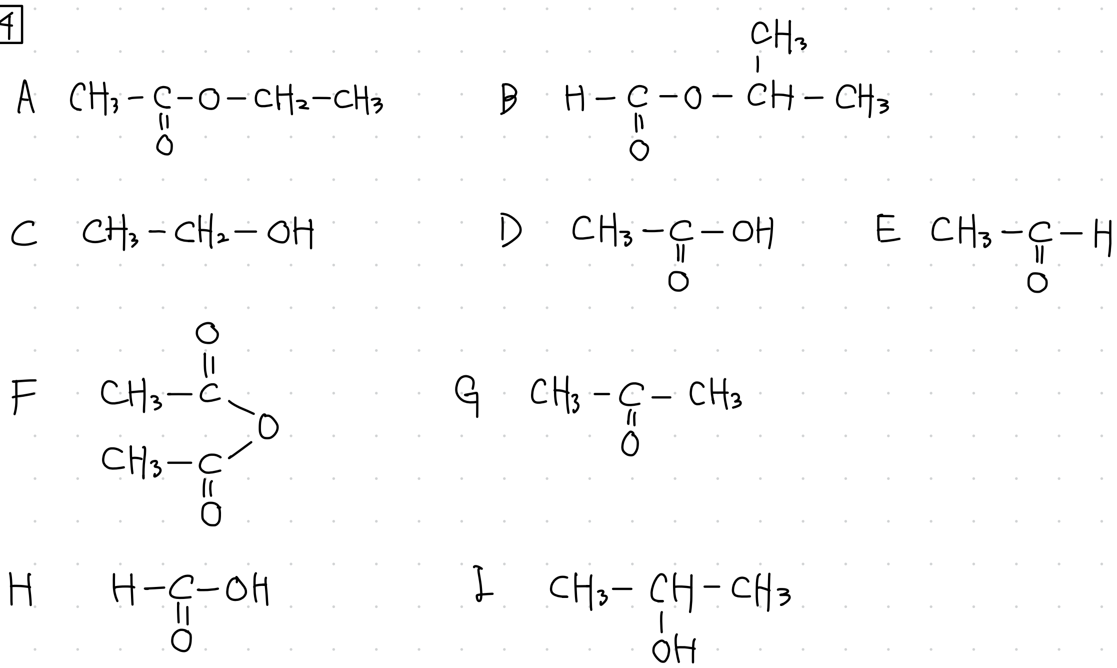
問 3 酢酸カルシウム

$(CH_3COO)_2Ca \rightarrow CH_3COCH_3 + CaCO_3$

問 4 4.4 g

3 − $\boxed{14}$

問1



問2,3 略

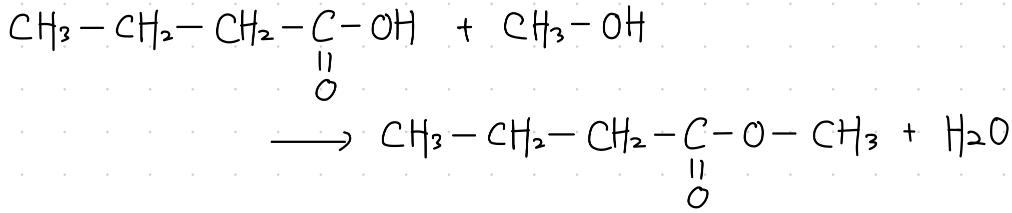
問4. アセトアルデヒド

問5 ジエチルエーテル

$2CH_3-CH_2-OH \longrightarrow CH_3-CH_2-O-CH_2-CH_3 + H_2O$

3－$\boxed{15}$

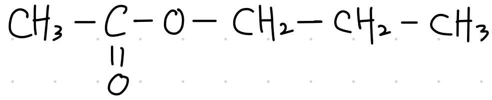
問1



エステル化の触媒

問2　$2C_5H_{10}O_2 + 13O_2 \longrightarrow 10CO_2 + 10H_2O$　　1.46×10^{-1} L

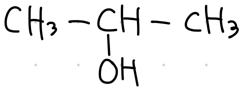
問3



B：酢酸

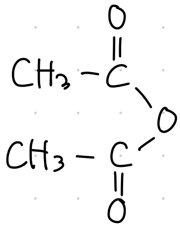
C：1－プロパノール

問4



$CH_3-CH_2-O-CH_2$

問5

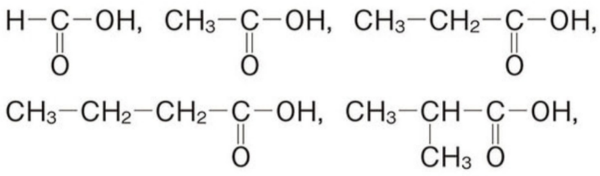


問6　ホルミル基

3－$\boxed{16}$

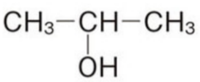
問1　10

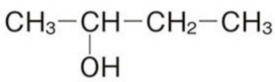
問2

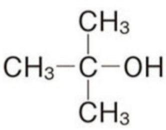


問3　ギ酸

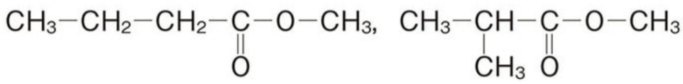
問4

A 

B 

D 

問5



問6